

Beliefs about Beliefs

Politicians' Rhetorical Ideology and Voter Perceptions

Daniel R. Daneri Perry J. Carter Will S. Schulz

Syracuse University NYU Abu Dhabi Stanford University

EPSA 2026 Prague

- Ideology scores are indispensable, but they are usually estimated from **elite behavior**.
- Roll-call scales capture legislative conflict under agenda control and party discipline (Poole and Rosenthal, 1985; Poole and Rosenthal, 2000).
- Donor-based scores capture location in a campaign-finance network (Bonica, 2014).
- Voters, by contrast, often encounter politicians through public language: posts, speeches, newsletters, ads, and interviews.

Empirical tension

Standard scores can have “little connection to voter perceptions” (The New York Times Editorial Board, 2025).

Question

Can we measure the ideology that politicians **convey rhetorically**, and does it better match how the public perceives them?

Behavioral vs. Rhetorical Ideology

Definition

Rhetorical ideology is the left-right position conveyed by a politician's public-facing language.

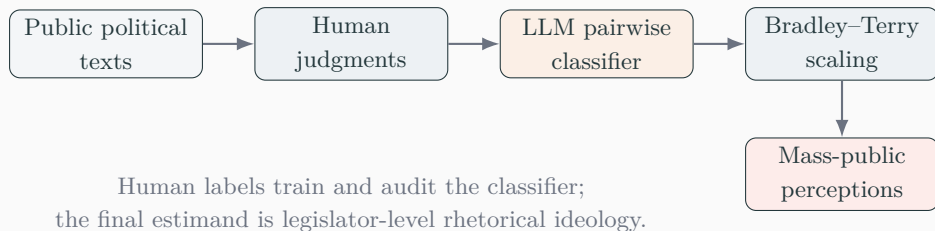
- It is not a claim about private belief.
- It is not a noisy proxy for true ideology.
- It is a political behavior in its own right.

Ideology	Empirical source
Behavioral	Votes, bills, ratings
Donor	Contributions and donor networks
	Tweets, speeches, newsletters
Rhetorical	
Perceived	Mass survey judgments

What is ideology?

We aim to avoid an expert definition of ideological content: ideology is whatever voters perceive it to be.

Research design in one slide



Core hypothesis

Rhetorical ideal points should be closer to public perceptions than DW-NOMINATE or CFscores, especially for less politically knowledgeable respondents.

Unit of analysis: pairwise comparisons

- Short political texts are hard to score in isolation, while automated vocabulary-based scaling performs weakly.
- A relative task is cognitively simpler:
which text is more conservative?
- Pairwise judgments preserve weak ordinal information and can be aggregated into a scale.

Annotation prompt

You will see two short political texts, A and B. Which text is more conservative? Conservative means right-wing or closer to the Republican Party; liberal means left-wing or closer to the Democratic Party.

Response options

A clearly / A arguably / B arguably / B clearly / not enough information.

Legislator-level contrast

For text pair (i, j) with authors $a(i)$ and $a(j)$:

$$\Pr(y_{ij} = 1 \mid \theta) = \sigma\{\theta_{a(i)} - \theta_{a(j)}\}.$$

- $y_{ij} = 1$ means text i is more conservative.
- Larger θ_ℓ means more conservative rhetoric.
- Tweets/newsletters are noisy draws from legislator rhetoric.

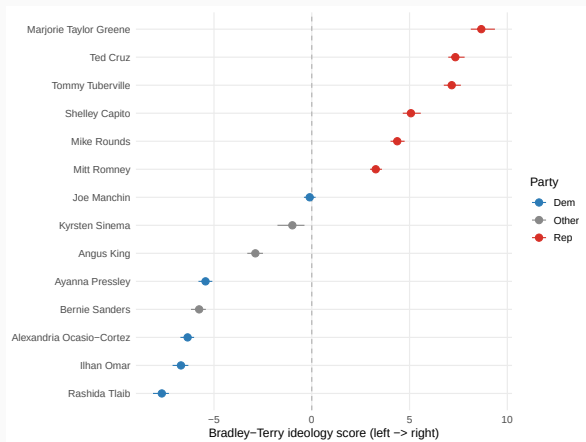
Regularized Bradley–Terry fit

$$\hat{\theta} = \arg \max_{\theta} \left\{ \ell(\theta; S_{BT}) - \frac{\lambda}{2} \|\theta\|_2^2 \right\}.$$

Orientation

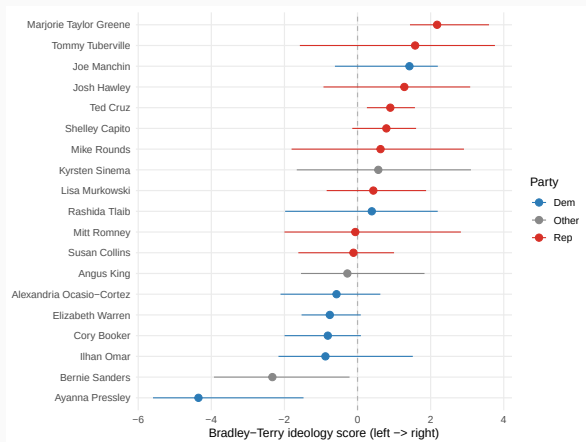
Scores are centered and oriented so higher values indicate more conservative rhetoric.

Metric	Hand-coded tweets	LLM-coded newsletters
Directional / valid comparisons	664	6,416
Dropped before scaling	194 non-directional	517 same-member
Member-level BT observations	664	5,899
Unordered member dyads	125	91
Politicians scaled	19	14
Connected components	1	1
Coders contributing	5	—



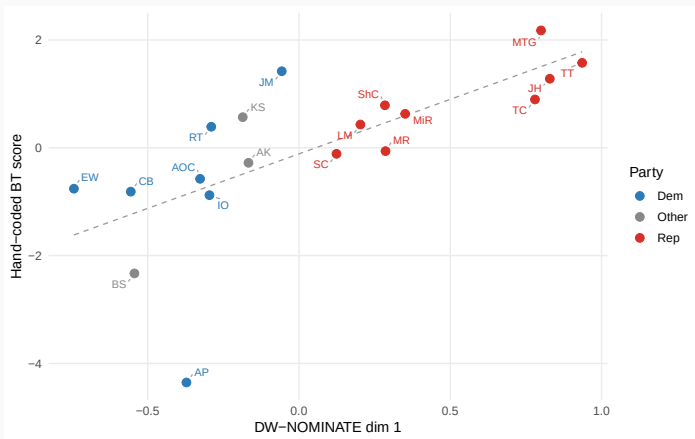
- Newsletter all-pairs coding yields much tighter intervals.
- The scale separates partisan rhetoric sharply.
- This shows throughput, but also motivates auditing.

Expert-coding validity



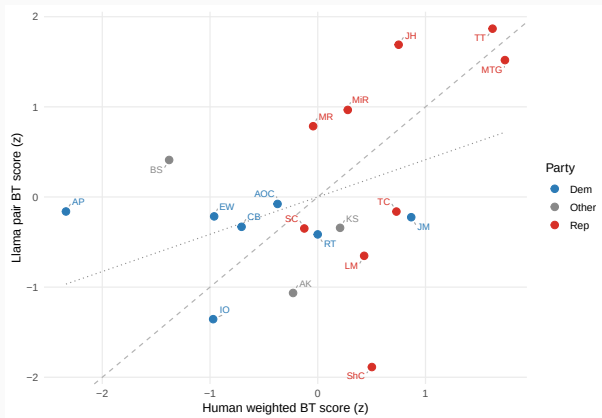
- Pilot estimates order prominent figures in familiar ways.
- Intervals are wide for some legislators, as expected with sparse human labels.

Convergent validity vs DW-NOMINATE



- Human-coded rhetoric broadly aligns with DW-NOMINATE.
- Large residuals are consistent with the idea that some legislators behave differently online from on the floor.

LLM vs. Human audit



- Broad structure, but important missed cases.
- Disagreements relate to conceptual uncertainty (tweets in this sample are closely matched).

Moving forwards: task-aware active learning

Problem

With N texts there are $O(N^2)$ possible pairs. Human labels are expensive, and most pairs are not equally useful.

Acquisition rule

$$\text{Score}_{\text{train}}(i, j) = U(i, j)V(i, j).$$

Classifier uncertainty U

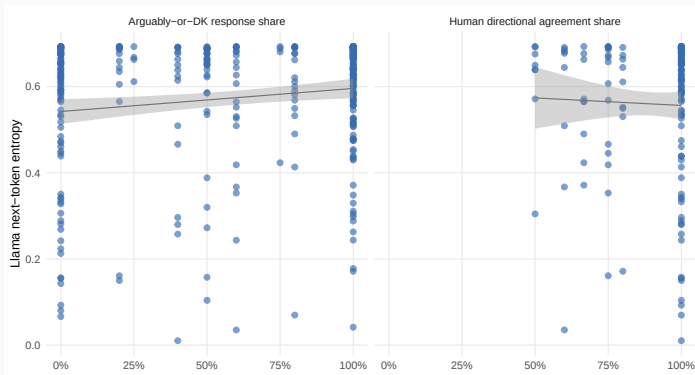
High when the model is close to indifferent between A and B .

Downstream value V

High when a label would reduce uncertainty about consequential legislator contrasts.

The key move is to select labels that train the classifier **and** improve the final rhetorical scale.

Model vs. human uncertainty



- Next-token entropy is useful but incomplete.
- Human uncertainty also reflects issue mismatch, ambiguous valence, and background knowledge.
- Ideally the two would correspond more closely → need for fine-tuning.

Mass-public perception scale

Survey respondents compare politicians or political texts; pairwise public judgments are aggregated into a perceived-ideology scale π_ℓ .

Primary comparison

$$\text{corr}(\pi, \hat{\theta}_{\text{rhet}}) > \text{corr}(\pi, \text{DW}),$$

$$\text{corr}(\pi, \hat{\theta}_{\text{rhet}}) > \text{corr}(\pi, \text{CF}).$$

- H1: rhetorical ideology better matches mass perceptions than DW-NOMINATE.
- H2: rhetorical ideology better matches mass perceptions than CFscores.
- H3: the advantage should be larger among lower-knowledge respondents.
- H4: higher text production may reveal larger rhetorical-behavioral gaps.

Questions for the audience

1. Quoted material. Should approving quote-tweets count as the legislator's rhetoric, be down-weighted, or be separated?
2. Political background knowledge. Should coders and models use broad contextual knowledge, or only text-internal cues?
3. Right endpoint. Should the conservative endpoint capture traditional conservatism, far-right populism, or a mixture?
4. Issue comparability. Should comparisons be aggressively topic-matched, or should cross-issue comparisons remain because voters collapse issues into a single scale?

Thank you

Comments welcome

Human annotation instrument

Task

You will be shown two short political texts, A and B. Which text is more conservative?

Definition

Conservative means right-wing or closer to the Republican Party. Liberal means left-wing or closer to the Democratic Party. Judge the text itself, including policy content, values, framing, tone, and ideological cues. Do not infer the answer from the author's identity.

Response options

A is clearly more conservative; A is arguably more conservative; B is arguably more conservative; B is clearly more conservative; not enough information / cannot determine.

LLM forced-choice prompt

System

You are an expert in American politics and political ideology.

User

I will give you two short political texts, A and B. Decide which text is more conservative on a U.S. liberal-conservative scale. Conservative means right-wing or closer to the Republican Party. Liberal means left-wing or closer to the Democratic Party. Judge the text, not the author. Answer with a single letter: A or B. Do not output anything else.

Token-probability audit

For models with logits, renormalize the first-token probabilities over A and B to obtain p_{ij}^{LLM} .

One-step downstream information value

Given current estimates $\hat{\theta}$, define

$$p_{\ell m} = \sigma(\hat{\theta}_{\ell} - \hat{\theta}_m), \quad w_{\ell m} = p_{\ell m}(1 - p_{\ell m}).$$

Let $C_t = (I_t + \lambda I)^{-1}$ be the approximate posterior covariance of the BT fit and

$$d_{\ell m} = (e_{\ell} - e_m)' C_t (e_{\ell} - e_m).$$

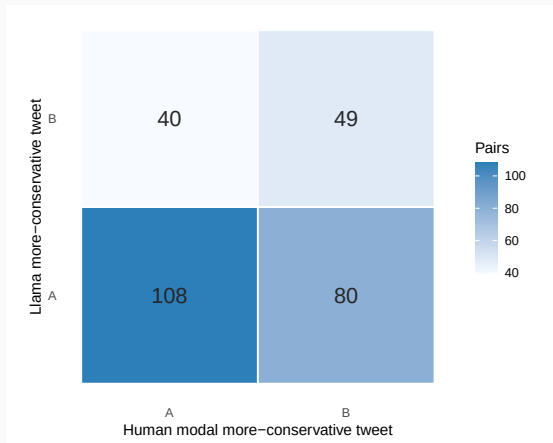
Then the one-step D-optimal value of another comparison between legislators ℓ and m is

$$V_{\text{leg}}(\ell, m) = \log\{1 + w_{\ell m} d_{\ell m}\}.$$

Intuition

Prioritize pairs that are locally hard to order and would reduce uncertainty about an important legislator contrast.

Pair-level Llama-human agreement



- Agreement is asymmetric across A/B presentation.
- Forced-choice labels can hide ambiguity in both human and model judgments.
- Swap-consistency and calibration checks should be part of production runs.

References

-  Bonica, Adam (2014). “Mapping the ideological marketplace”. In: American Journal of Political Science 58.2, pp. 367–386.
-  Poole, Keith T and Howard Rosenthal (1985). “A spatial model for legislative roll call analysis”. In: American Journal of Political Science 29.2, pp. 357–384.
-  — (2000). Congress: A political-economic history of roll call voting. Oxford University Press, USA.
-  The New York Times Editorial Board (2025). The partisans are wrong: Moving to the center is the way to win. The New York Times. October 20.